

3 Method

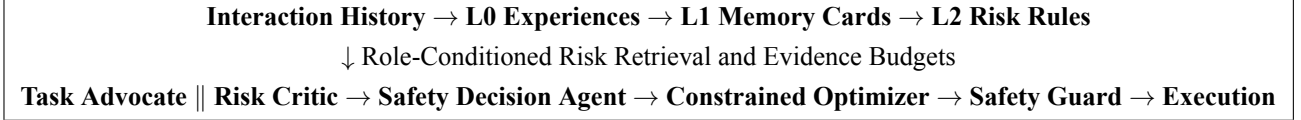


Fig. 1: Method overview. Hierarchical memory compresses long-term history while preserving provenance; each role receives distinct risk evidence; deterministic modules retain execution authority.

3.1 System Overview

We study high-level semantic action selection for an indoor mobile robot in a partially observable environment. At time t , the planning input $\mathcal{X}_t = (g_t, r_t, e_t, \mathcal{A}_t)$ contains task goal and progress, robot state, environment observation, and a finite admissible action set. The system converts long-term history $\mathcal{H}_t$ into bounded, traceable risk evidence.

Figure 1 summarizes the pipeline, whose mathematical thread is

$$E_i \rightarrow z_i \rightarrow q_i \rightarrow M_i \rightarrow (Q_j, c_j) \rightarrow \mathcal{B}_r^* \rightarrow a_t^*. \quad (1)$$

An L0 experience yields a risk state and single-case evidence, which form an L1 card; related cards produce L2 rules, and role-conditioned retrieval supplies evidence for action selection. The Task Advocate, Risk Critic, and Safety Decision Agent return structured recommendations and evidence IDs. A deterministic optimizer retains action authority, followed by an independent Safety Guard.

3.2 L0–L1–L2 Hierarchical Risk Memory

The hierarchy contains L0 experiences $\mathcal{E}_t$, L1 cards $\mathcal{M}_t$, and L2 rules $\mathcal{Q}_t$. Stable IDs connect their provenance, allowing each abstract rule to be traced to cards and source trajectories.

3.2.1 L0: Embodied Risk Experiences

Each L0 experience E_i records the task, robot and environment state, action A_i , outcome O_i , failure cause, risk type and severity $\sigma_i \in \{1, \dots, 5\}$, decision, and trajectory provenance. Its immutable `experience_id` links sensor logs, control records, or human labels. Ingestion validates required fields, outcome values, severity, and ID uniqueness.

A deterministic projection ψ extracts risk state $z_i = [1 - \bar{c}_i, 1 - \bar{d}_i, 1 - p_i, 1 - b_i, h_i]$. Here $\bar{c}_i$ and $\bar{d}_i$ are normalized passage clearance and obstacle distance, p_i is observation confidence, b_i is battery ratio, and h_i encodes robot type, obstacle state, and task. The first four dimensions share a direction: larger values indicate stronger risk factors. L0 retains the full trajectory for audit.

3.2.2 L1: Risk Memory Cards

L1 converts the outcome into single-case evidence $q_i = (\sigma_i/5)\omega(O_i)\gamma_i$, where $\gamma_i \in [0, 1]$ is experience reliability and outcome factor $\omega(O_i)$ equals 1, 0.8, and 0.05 for failure, aborted, and success. The value q_i is relative risk evidence, not collision probability. An explicit quality score supplies γ_i when available; otherwise it is estimated from smoothed success/failure statistics.

Substituting z_i and q_i into the compression map derives the

L1 card:

$$\begin{aligned} M_i &= C(E_i) \\ &= \text{Pack}(\psi(E_i), A_i, O_i, F_i, B_i, \\ &\quad \frac{\sigma_i}{5}\omega(O_i)\gamma_i, \pi_i) \\ &= \text{Pack}([1 - \bar{c}_i, 1 - \bar{d}_i, 1 - p_i, 1 - b_i, h_i], \\ &\quad A_i, O_i, F_i, B_i, \frac{\sigma_i}{5}\omega(O_i)\gamma_i, \pi_i). \end{aligned} \quad (2)$$

Here F_i is the failure cause, B_i contains mitigations and stopping conditions, and π_i stores the `memory_id`, `experience_id`, and source span. The three lines of Eq. (2) respectively show the source experience, substitution of risk state and outcome evidence, and the resulting card, making the L0-to-L1 construction explicit.

3.2.3 L2: Risk Rules and Cognitive Evolution

Cards with similar z_i , risk type, and action semantics form a pattern set $\mathcal{I}_j$. An aggregator extracts trigger conditions, recommended and prohibited actions, mitigations, and source IDs to form rule Q_j . Let n_j^+ , n_j^- , and n_j^0 count mitigation successes, risk failures, and inconclusive aborts; the first two are decisive evidence. With a Beta(α, β) prior, rule evidence reliability is

$$\begin{aligned} (Q_j, c_j) &= \mathcal{A}(\{M_i\}_{i \in \mathcal{I}_j}), \\ c_j &= \mathbb{E}[\theta_j | \mathcal{I}_j] = \frac{n_j^+ + n_j^- + \alpha}{n_j^+ + n_j^- + n_j^0 + \alpha + \beta}. \end{aligned} \quad (3)$$

Here c_j measures whether a pattern has decisive empirical support; it is not the probability of a hazard. A *candidate* becomes *active* after meeting provenance-count, cross-scene-consistency, and reliability thresholds. Conflicting or stale evidence can revise its trigger or mark it *retired*. Each transition preserves prior versions and source IDs, maintaining the chain $Q_j \rightarrow M_i \rightarrow E_i$.

At runtime, only *active* rules are matched. The system retains actions recommended by all matched rules and prohibited by none. If the intersection is empty, a deterministic conflict gate stops execution and requests human review.

3.3 Role-Conditioned Risk-Aware Retrieval

Given $\mathcal{X}_t$, each card produces $\phi_i = [\text{sim}, \text{sev}, \text{rel}, \text{rec}, \text{fit}] \in [0, 1]^5$. Its first three components come from the risk state, outcome evidence, and card quality in Eq. (2); the others encode recency and role fit. Role weights $w^{(r)}$ make the Advocate emphasize successes and mitigations, the Critic failures, severe hazards, and stopping conditions, and the Decision role balance positive and negative cases.

Retrieval jointly optimizes role relevance, case diversity,

and context size:

$$\begin{aligned} \mathcal{B}_r^* = \arg \max_{\mathcal{B} \subseteq \mathcal{M}_t} & \sum_{M_i \in \mathcal{B}} (w^{(r)})^\top \phi_i - \lambda \sum_{\substack{M_i, M_j \in \mathcal{B} \\ i < j}} d(i, j) \\ \text{s.t. } |\mathcal{B}| \leq K, & \sum_{M_i \in \mathcal{B}} \hat{\tau}(M_i) \leq B_r. \end{aligned} \quad (4)$$

$d(i, j)$ measures redundancy in risk type, outcome, and mitigation, while $\hat{\tau}$ conservatively estimates card tokens. Ordered as (sim, sev, rel, rec, fit), the reproducible baseline uses $w^{(\text{adv})} = (.35, .05, .20, .05, .35)$, $w^{(\text{crit})} = (.30, .25, .15, .05, .25)$, and $w^{(\text{dec})} = (.35, .20, .20, .10, .15)$; ablations isolate their effects. Algorithm 1 greedily approximates Eq. (4).

The bundle contains the role, retrieval version, budget usage, L1 cards, score decomposition, match rationales, and source IDs, plus matched L2 rules. Agent outputs declare used evidence in `cited_experience_ids` and `cited_rule_ids`; IDs absent from the bundle are rejected. These records identify what was retrieved, why, and what the agent cited.

Let $\mathcal{D}_t$ denote the three role reports. The deterministic optimizer first constructs feasible set $\mathcal{F}_t$ from hard constraints and matched L2 prohibitions, then uses the Decision bundle to solve

$$\begin{aligned} a_t^* &= \arg \max_{a \in \mathcal{F}_t} U(a), \\ R(a \mid \mathcal{B}_{\text{dec}}^*) &\leq \rho, \\ Z(a \mid \mathcal{D}_t, \mathcal{B}_{\text{dec}}^*) &\leq \eta. \end{aligned} \quad (5)$$

R combines scene risk with historical evidence from Eq. (2); Z combines observation confidence, memory coverage, and agent disagreement. Both are relative scores. Rule conflict, invalid agent output, an empty feasible set, or Safety Guard rejection triggers `safe_stop`, `observe_again`, or human review. Equation (1) therefore closes memory construction into executable decision making.

Algorithm 1: Role-Conditioned Risk Retrieval

Require: State $\mathcal{X}_t$, roles $\mathcal{R}$, L0 store $\mathcal{E}$, budgets $\{B_r\}$, limit K

Ensure: Role-specific evidence bundles $\{\mathcal{B}_r\}_{r \in \mathcal{R}}$

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1: Validate  $\mathcal{X}_t$ ,  $\mathcal{R}$ , and  $\mathcal{E}$ 
2: Build each  $M_i = C(E_i)$  by Eq. (2)
3: for each role  $r \in \mathcal{R}$  do
4:   Compute all  $\phi_i$ ; initialize candidates  $\mathcal{C}_r$  and  $\mathcal{B}_r \leftarrow \emptyset$ 
5:   while  $\mathcal{C}_r \neq \emptyset$  and  $|\mathcal{B}_r| < K$  do
6:     Select  $M_{i^*}$  by marginal gain in (4)
7:     Remove  $M_{i^*}$  from  $\mathcal{C}_r$ 
8:     if  $\hat{\tau}(\mathcal{B}_r) + \hat{\tau}(M_{i^*}) \leq B_r$  then
9:        $\mathcal{B}_r \leftarrow \mathcal{B}_r \cup \{M_{i^*}\}$ 
10:    end if
11:  end while
12:  Append matched active L2 rules, scores, rationales, and
    source IDs to  $\mathcal{B}_r$ 
13: end for
14: return  $\{\mathcal{B}_r\}_{r \in \mathcal{R}}$ 

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